Skin Cancer Detection Using Deep Learning: A Comparative Analysis and a Multi-Model CNN–Transformer Ensemble with External Validation

Abhishek Patil, Jay Tripathi, Paras Shinkar, and Salman Sheikh

Abstract

Early detection is the single most important factor in surviving skin cancer, yet access to dermatologists is limited and visual diagnosis of melanoma, basal cell carcinoma (BCC), and squamous cell carcinoma (SCC) is error-prone even for experts. Deep learning has produced many image classifiers for this task, but their published accuracies are difficult to compare because studies differ in datasets, class definitions, data splits, and metrics. This paper makes three contributions. First, it presents a structured literature review of 40+ studies covering convolutional neural networks (CNNs), vision transformers, hybrid and ensemble models, dermatology foundation models, prospective clinical trials, explainability, and skin-tone fairness. Second, it performs a comparative analysis of published results for classical machine learning, VGG, ResNet, DenseNet, MobileNet, EfficientNet, ConvNeXt, ViT, Swin, hybrid CNN–transformer, ensemble, and foundation models on the HAM10000, ISIC 2019, and ISIC 2020 benchmarks, and for U-Net-family segmentation on ISIC 2018, separating random from patient-disjoint and official-test protocols. The analysis shows that split leakage and dataset shift can account for differences larger than those between architectures. Third, guided by this evidence, it trains and evaluates a multi-model ensemble that combines two CNNs (EfficientNetV2-S and ConvNeXt-T) and a vision transformer (Swin-T) by temperature-calibrated soft voting. The models are trained with focal loss on ISIC 2019 (25,331 dermoscopic images, eight classes) using a de-duplicated, lesion-grouped split, and are tested on a held-out internal set of 3,596 images and on the external smartphone dataset PAD-UFES-20 (2,298 images). On the internal test set, equal-weight soft voting achieved 77.95% accuracy, 62.99% balanced accuracy, a macro-AUC of 0.950, and 95.4% malignant sensitivity, exceeding the best single model (ConvNeXt-T: 76.89%, 60.21%, and 0.936); validation-weighted voting achieved 76.89% accuracy and 61.65% balanced accuracy. On external smartphone images, accuracy fell below 38% for every model, quantifying the domain-shift problem identified in the review, yet model combination still raised the malignant-versus-benign AUC from at most 0.844 to 0.873.

Index Terms

Skin cancer, melanoma, deep learning, convolutional neural network, vision transformer, ensemble learning, lesion segmentation, dermoscopy, explainable AI, fairness.

I. Introduction

SKIN cancer is among the most frequently diagnosed cancers worldwide. Melanoma causes most skin-cancer deaths, while the keratinocyte cancers—BCC and SCC—are far more common and impose a large treatment burden. Prognosis depends strongly on the stage at which a lesion is found, so a tool that helps a non-specialist decide quickly whether a lesion is suspicious has clear clinical value.

Visual assessment, with or without a dermatoscope, is the first step in diagnosis. It is subjective, and its accuracy drops sharply outside specialist clinics. Deep learning changed the field when a single CNN matched board-certified dermatologists on biopsy-proven clinical images [1]. Reader studies later reported comparable or better performance than dermatologist panels on dermoscopic images [2]–[4]. Meta-analytic evidence now indicates that AI reaches a pooled sensitivity of 87.0% and specificity of 77.1%, above the corresponding values for clinicians as a group [5].

Two problems still limit progress. *First*, reported results are hard to compare. Papers on the popular HAM10000 set [6] often split images at random, even though many lesions contribute several near-identical photographs, so

The authors are with the Department of Computer Engineering, Shah & Anchor Kutchhi Engineering College (Autonomous, Affiliated to University of Mumbai), Mumbai, Maharashtra, India. This work was carried out with the guidance of Dr. Bhakti Sonavane.

the same lesion can appear in both training and test sets. When duplicates are removed from the ISIC archives, the best area under the ROC curve (AUC) on the ISIC 2020 test set falls to 0.80 [7]. The ISIC 2019 winning ensemble lost about ten points of balanced accuracy between cross-validation and the official hidden test set [8]. *Second*, benchmark success has not reliably transferred to patients. In prospective trials, AI tuned for high sensitivity often has low specificity [9], [10], and a smartphone algorithm that led experimental benchmarks was inferior to specialists in the clinic [11]. Public datasets also under-represent darker skin [12], and model accuracy falls on these patients [13], [14].

This paper addresses both problems. Its contributions are:

1) A thematic **literature review** of more than 40 studies (Section II) that identifies six concrete research gaps.
2) A curated **dataset analysis** (Section III) of eleven public skin-cancer datasets with verified image counts, class balance, and recommended roles, including the recent SLICE-3D and MILK10k collections.
3) A **proposed methodology** (Section IV): a hybrid CNN–transformer ensemble with imbalance-aware training, probability calibration, and a sensitivity-driven referral rule, designed so that segmentation, metadata fusion, and explanations can be added as modules.
4) A leakage-free **experimental design** (Section V) with de-duplication, lesion-grouped splitting, and an external test set.
5) **Measured results** (Section VI) of the three backbones and their combination on internal and external data, with training curves, confusion matrices, ROC curves, confidence intervals, and a skin-type breakdown.
6) A **comparative analysis of multiple algorithms** (Section VII) that places published results and our own side by side, grouped by comparable protocol, together with graphs and a qualitative trade-off analysis.

The remainder of the paper follows this order: literature review, datasets, proposed methodology, experimental design, experimental results, comparative analysis, discussion, and conclusion.

II. Literature Review

We searched IEEE Xplore, PubMed, Scopus, and arXiv for studies published from 2015 to 2026 using combinations of the terms *skin cancer*, *melanoma*, *skin lesion*, *deep learning*, *CNN*, *transformer*, *segmentation*, and *dermoscopy*. We gave priority to peer-reviewed journal papers, challenge reports, and prospective clinical studies, and verified every bibliographic record against its DOI. Earlier surveys [15]–[17] catalogue architectures but rarely compare them under a common protocol; this review therefore emphasizes evaluation conditions alongside reported numbers. Table I summarizes representative studies; the paragraphs below organize the literature into seven themes.

A. Early Deep Learning and Reader Studies

Esteva *et al.* [1] fine-tuned an Inception network on about 129,000 clinical images and reached dermatologist-level performance on two binary tasks. Haenssle *et al.* [2] and Brinker *et al.* [3] compared CNNs with 58 and 157 dermatologists on dermoscopic melanoma images and found the networks competitive or superior. In the ISIC 2018 reader study, the top three algorithms averaged 25.43 correct answers per 30-image batch against 18.78 for experts, although their advantage shrank on images from outside the training distribution [4]. These studies established feasibility but used curated, retrospective images.

B. Convolutional Neural Networks

Transfer learning from ImageNet with standard CNNs—VGG [18], ResNet [19], Inception [20], DenseNet [21], Xception [22], and MobileNet [23]—is the most common approach. On HAM10000, a MobileNet reached 83.1% accuracy with a low computational cost [24]. A comparison of five CNNs found ResNeXt-101 best at 93.2% [25], and adding soft attention to InceptionResNetV2 raised accuracy to 93.4% [26]. Among EfficientNet variants [27], B4 gave the best trade-off at 87.9% accuracy [28]. Shetty *et al.* [29] showed that a small custom CNN clearly outperforms classical baselines such as a support vector machine (SVM) or random forest trained on the same data.

C. Transformers, Hybrids, and Ensembles

Vision transformers (ViT) [30] model long-range relations between image patches. A modified ViT, SkinTrans, reported 94.3% on HAM10000 [31]. On the harder eight-class ISIC 2019 set, a Swin Transformer [32] reached a balanced accuracy of 82.3% [33], and an ensemble of four ConvNeXt [34] models from the same group raised the mean sensitivity to 84.2% [35]. Hybrid designs pair a CNN, which captures local texture, with a transformer, which captures global context. Nie *et al.* [36] combined a ResNet stem with a ViT and focal loss, and Karthik *et al.* [37] fused Swin with a DenseNet attention branch (94.2% on HAM10000). MedFusionNet fuses ConvNeXt and ViT features with attention [38]. Ensembles dominate challenge leaderboards: Gessert *et al.* [8] won ISIC 2019 with multi-resolution EfficientNets plus a metadata branch, and an EfficientNet ensemble won the SIIM-ISIC 2020 melanoma challenge [39].

D. Lesion Segmentation

Segmentation isolates the lesion from surrounding skin, hair, and rulers. U-Net [40] remains the reference model; attention gates [41], transformer encoders (TransUNet) [42], and channel-wise transformer skip connections (UCTransNet) [43] are its main successors. A survey of 177 papers [44] identifies artefacts and inconsistent annotation as the main open problems, and a uniform re-evaluation of U-Net variants on ISIC 2018 [45] provides directly comparable Dice scores (Section VII).

E. Class Imbalance and Synthetic Data

Malignant lesions are the minority in almost every dataset (Fig. 2). Focal loss [46] down-weights easy examples. Under an identical backbone and split, it raised overall accuracy from 74.2% to 83.7% [36]. Generative models provide an alternative route: an improved DCGAN reached high accuracy on a binary ISIC 2017 task [47], synthetic StyleGAN2 melanomas raised melanoma AUC from 0.925 to 0.945 [48], and Stable Diffusion has also been used to synthesize training lesions [49]. Evidence that synthetic data helps on *external* test sets is still limited.

F. Explainability and Foundation Models

A systematic review of explainable AI (XAI) for skin cancer found that only 3 of 37 studies measured how explanations affect clinicians [50]. Grad-CAM [51] is the default tool, whereas dermoscopy-feature-based explanations measurably increased dermatologists' trust [52]. Foundation models are the newest trend. PanDerm was pretrained on more than two million images and set the state of the art on 28 benchmarks [53]. MONET links images to medical concepts for model auditing [54], and SkinGPT-4 adds a conversational interface [55]. These models are powerful but computationally heavy. Lightweight alternatives such as HI-MViT [56] and SkinNet-14 [57] target mobile deployment.

G. Clinical Validation and Fairness

Prospective studies give a more sober picture. The PROVE-AI study reached 96.8% sensitivity but only 37.4% specificity [9], the DERM device detected all melanomas but had low specificity for BCC and SCC [10], and a multicentre German study found AI more sensitive but less specific than dermatologists [58]. Only two of 272 primary-care studies used low-prevalence data [59]. Regarding equity, skin type is recorded for only 2.1% of public images [12]. Models degrade on the diverse DDI set [13], and AI assistance can widen the accuracy gap between light and dark skin [14].

H. Research Gaps

The review reveals six gaps that motivate this work:

G1 *Incomparable evaluation:* random image-level splits, varying class sets, and accuracy-only reporting inflate and obscure results [7].

G2 *No external validation:* most architecture papers test only on data from the training distribution [37], [38].

G3 *Imbalance and non-melanoma cancers:* BCC, SCC, and actinic keratosis receive less attention than melanoma [59].

G4 *Skin-tone bias:* performance is rarely reported per Fitzpatrick type [12], [13].

TABLE I
SUMMARY OF REVIEWED LITERATURE ON DEEP LEARNING FOR SKIN CANCER DETECTION (RESULTS AS REPORTED BY THE AUTHORS).

Study	Year	Method	Dataset	Reported result	Limitation / gap
Esteva *et al.* [1]	2017	Inception CNN, transfer learning	129k clinical images	On par with 21 dermatologists	Private data; binary tasks
Tschandl *et al.* [4]	2019	ISIC 2018 algorithms vs. 511 readers	ISIC 2018 test	Top-3 AI: 25.4 vs. 18.8 correct/30	Advantage shrinks out of distribution
Chaturvedi *et al.* [25]	2020	Five ImageNet CNNs	HAM10000	Best (ResNeXt-101) 93.2% acc.	Random split; no external test
Gessert *et al.* [8]	2020	Multi-res. EfficientNet ensemble + metadata	ISIC 2019	BA 74.2% (CV), 63.6% (official test)	Heavy ensemble; large CV–test gap
Datta *et al.* [26]	2021	InceptionResNetV2 + soft attention	HAM10000	93.4% acc., AUC 0.984	Resampled random split
Ali *et al.* [28]	2022	EfficientNet B0–B7	HAM10000	B4: 87.9% acc., F1 0.87	Single dataset
Xin *et al.* [31]	2022	Improved ViT (SkinTrans)	HAM10000	94.3% acc.	Split not detailed
Cassidy *et al.* [7]	2022	Duplicate audit of ISIC archives, 19 CNNs	ISIC 2016–2020	Best AUC 0.80 after de-duplication	Shows benchmark inflation
Ayas [33]	2023	Swin Transformer	ISIC 2019 (8 cl.)	BA 82.3%, spec. 97.9%	Single model; no metadata
Nie *et al.* [36]	2023	ResNet + ViT hybrid, focal loss	ISIC 2018	89.5% acc.; macro-F1 81.1%	No external validation
Ding *et al.* [56]	2023	Lightweight MobileViT (HI-MViT)	ISIC 2018/2017, PH2	93.2% acc., AUC 0.977	No on-device latency reported
Marchetti *et al.* [9]	2023	Prospective open-source ensemble	603 lesions, MSKCC	Sens. 96.8%, spec. 37.4%	Low specificity; 96% white cohort
Menzies *et al.* [11]	2023	Smartphone AI, prospective trial	172 lesions, AU/AT	7-class AI equal to specialists	ISIC AI inferior in clinic
Baykal Kablan & Ayas [35]	2024	ConvNeXt ensemble	ISIC 2019 (8 cl.)	Sens. 84.2%, spec. 97.9%	Same-lab split; no external test
Karthik *et al.* [37]	2024	Swin + DenseNet attention hybrid	HAM10000	94.2% acc., F1 96.6%	One dataset only
Heinlein *et al.* [58]	2024	Prospective multicentre AI	8 German hospitals	BA 0.798 vs. 0.781 (dermatol.)	Specificity 0.67 vs. 0.83
Groh *et al.* [14]	2024	AI-assisted diagnosis experiment	364 images, 848 physicians	AI aid >33% accuracy gain	Widens skin-tone gap
Salinas *et al.* [5]	2024	Meta-analysis, AI vs. clinicians	53 studies	AI sens. 87.0%, spec. 77.1%	Mostly retrospective
Chanda *et al.* [52]	2024	Dermoscopy-feature XAI	Reader study	Higher clinician trust	Melanoma vs. nevus only
Yan *et al.* [53]	2025	PanDerm foundation model	2M+ images, 28 tasks	SOTA on 28 benchmarks	Very high compute cost
Naeem *et al.* [38]	2025	ConvNeXt + ViT attention fusion	HAM10000, ISIC 2019	98.8% / 97.9% acc.	Benchmark-only evaluation
Lin *et al.* [48]	2026	StyleGAN2 synthetic melanoma	ISIC 2018/2020	Melanoma AUC 0.925→0.945	Melanoma class only

G5 *Limited explainability:* saliency maps are added but seldom evaluated [50].

G6 *Accuracy versus deployability:* strong foundation models are heavy, while light models lack external tests [53], [56].

III. DATASETS

Choosing datasets is as important as choosing an architecture. We reviewed eleven public datasets (Table II). The experiments in this paper use the **two** that best fit the task: ISIC 2019 for training and internal testing, and PAD-UFES-20 as an external test set. Together they cover all three major skin cancers, two imaging modalities, and two continents, without overlapping images.

Training and internal test data: ISIC 2019. ISIC 2019 is the largest multi-class dermoscopic set, with 25,331 training images in eight classes, including melanoma (MEL), BCC, SCC, and actinic keratosis (AK) (Fig. 1). It already contains HAM10000 [6] and BCN20000 [60], so these are *not* added again. We split it into training (18,076 images), validation (3,659), and internal test (3,596) sets as described in Section V.

External test data: PAD-UFES-20. PAD-UFES-20 [61] contains 2,298 smartphone images collected in Brazil, with biopsy-proven cancers and the patient's Fitzpatrick skin type. It is never used for training. Its six diagnoses

TABLE II
PUBLIC SKIN-CANCER DATASETS REVIEWED (COUNTS VERIFIED AGAINST OFFICIAL LABEL FILES OR DATA PAPERS) AND THEIR ROLE IN THIS WORK.

Dataset	Year	Images	Modality	Classes (malignant in bold)	Metadata	Role in this work
HAM10000 [6]	2018	10,015	Dermoscopic	7: **MEL**, **BCC**, **AKIEC**, NV, BKL, DF, VASC	Age, sex, site	Included via ISIC 2019
ISIC 2017 [69]	2017	2,750	Dermoscopic	3: **MEL**, SK, NV	Age, sex	Background / literature
ISIC 2018 [64]	2018	2,594 (T1)	Dermoscopic	Masks (Task 1); 7 classes (Task 3)	–	Future: segmentation
ISIC 2019 [8]	2019	25,331	Dermoscopic	8: **MEL**, **BCC**, **SCC**, **AK**, NV, BKL, DF, VASC	Age, sex, site	**Used: training + internal test**
BCN20000 [60]	2024	18,946	Dermoscopic	8 (as ISIC 2019) + OOD	Age, sex, site	Included via ISIC 2019
ISIC 2020 [62]	2020	33,126	Dermoscopic	Binary: **MEL** (1.8%) vs. benign	Patient ID, age, sex, site	Future: melanoma training
SLICE-3D [66]	2024	401,059	3D-TBP crops	Binary: **malignant** (0.1%) vs. benign	Age, sex, site, TBP features	Future triage work
MILK10k [63]	2026	10,480	Clinical + dermoscopic	11 incl. **BCC**, **SCC**, **MEL**, **AKIEC**	Age, sex, site, skin tone	Future: external test
PH2 [65]	2013	200	Dermoscopic	3: **MEL**, atypical NV, common NV	Masks, criteria	Future: segmentation test
PAD-UFES-20 [61]	2020	2,298	Smartphone	6: **BCC**, **SCC**, **MEL**, ACK, SEK, NEV	21 fields incl. Fitzpatrick	**Used: external test**
Fitzpatrick 17k [70]	2021	16,577	Clinical	114 conditions; 3-way incl. **malignant**	Fitzpatrick I–VI	Optional fairness analysis
DDI [13]	2022	656	Clinical	Binary: **malignant** vs. benign	Fitzpatrick groups	Future: fairness test

map onto ISIC 2019 classes (ACK→AK, NEV→NV, SEK→BKL; BCC, SCC, and MEL unchanged), so it tests whether a dermoscopy-trained model transfers to ordinary phone photographs.

Datasets for the next stage. ISIC 2020 [62] (33,126 images, 1.8% melanoma) can add realistic-prevalence melanoma training data. MILK10k [63], the ISIC 2025 benchmark (5,240 lesions, 11 classes), and DDI [13], which is balanced across Fitzpatrick groups, are planned as further external and fairness test sets. ISIC 2018 Task 1 [64] and PH2 [65] provide lesion masks for the segmentation module. SLICE-3D [66] (401,059 crops, 0.1% malignant), the basis of the ISIC 2024 triage challenge [67], is reserved for future triage experiments. Derm7pt [68], which annotates the seven-point dermoscopic checklist, is a candidate for evaluating concept-level explanations. Fig. 2 shows how strongly malignant prevalence varies between datasets. This variation is a key reason why a single accuracy number is misleading.

IV. PROPOSED METHODOLOGY

The proposed system, shown in Fig. 3, answers the gaps of Section II with evidence-backed components rather than an untested new architecture. It contains six stages: preprocessing, lesion segmentation, a three-backbone hybrid ensemble, metadata fusion, calibrated decision making, and explanation. Section IV-A states exactly which of these stages are evaluated in the experiments of this paper.

A. Implemented Configuration

The experiments reported in Section VI evaluate the core of the framework: the three-backbone ensemble (Section IV-D), imbalance-aware training (Section IV-F), and the calibrated soft vote with its referral rule (Section IV-G), applied to whole lesion images. Lesion segmentation, metadata fusion, and Grad-CAM (Sections IV-C, IV-E, and

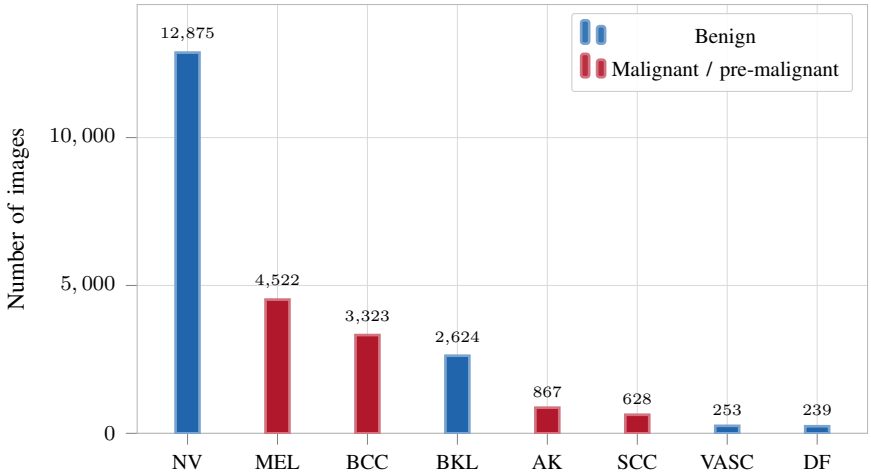


Fig. 1. Class distribution of the ISIC 2019 training set (25,331 images). The largest class (NV) is 54× the smallest (DF), and SCC has only 628 images, which motivates imbalance-aware training.

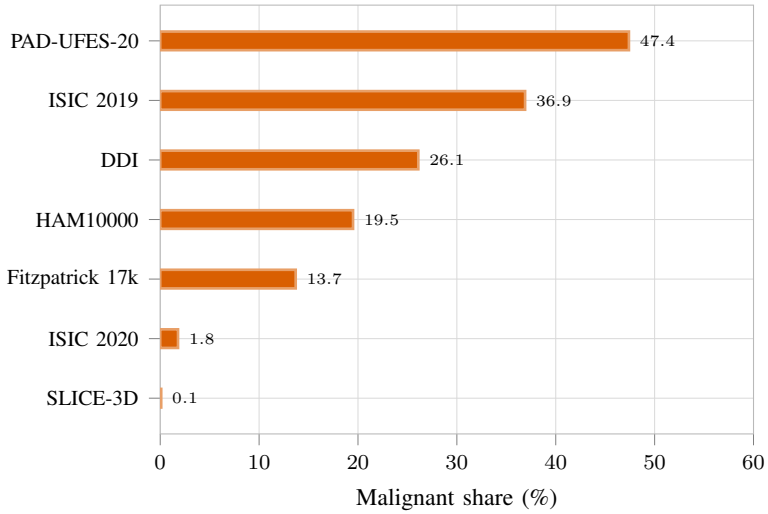


Fig. 2. Share of malignant images per dataset, computed from the official label files (HAM10000 and ISIC 2019 include the AKIEC/AK class; PAD-UFES-20 counts BCC, SCC, and MEL). Prevalence ranges from 0.1% to 47%, so accuracy alone is not comparable across datasets.

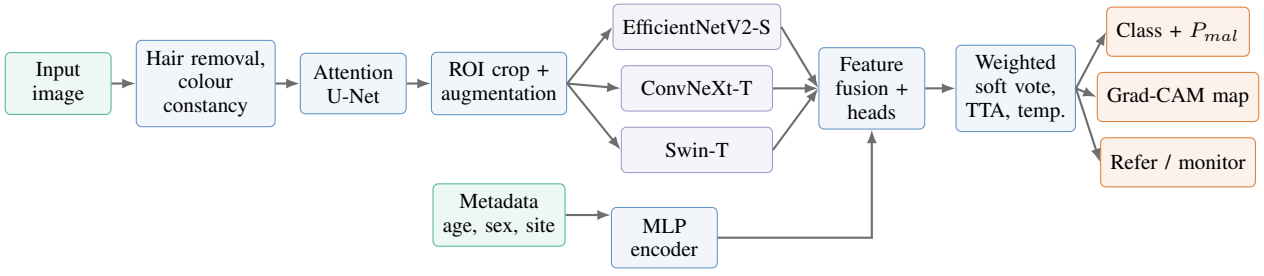


Fig. 3. Proposed hybrid ensemble framework. The lesion is classified by three complementary backbones (two CNNs, one transformer) whose calibrated outputs are combined by soft voting into class probabilities and a referral recommendation. The experiments in Section VI evaluate the three-backbone ensemble with calibration and the referral rule on whole images; segmentation, metadata fusion, and Grad-CAM are modules for the next stage.

IV-H) are designed as plug-in modules for the next stage, so that the contribution of each can later be measured against the baseline reported here.

B. Preprocessing

All images are first resized so that their shorter side is 256 pixels. During training, a random resized crop (scale 0.5–1.0) of 224×224 pixels is taken; at test time the image is resized to 224 pixels and centre-cropped. Hair and ruler artefact removal (morphological black-hat filtering with inpainting) and Shades-of-Gray colour constancy belong to the full framework but were not applied in the reported experiments.

C. Lesion Segmentation (Next Stage)

An attention U-Net [41] with an EfficientNet encoder will be trained on ISIC 2018 Task 1. Attention gates in the skip connections suppress background skin. Training minimizes the sum of the Dice loss and binary cross-entropy:

$$\begin{aligned}\mathcal{L}_{seg} = 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon} \\ - \frac{1}{N}\sum_i \big[g_i \log p_i + (1 - g_i)\log(1 - p_i)\big],\end{aligned} \tag{1}$$

where p_i is the predicted foreground probability and g_i the ground-truth label of pixel i. The mask defines a region of interest (ROI) that is cropped with a 15% margin so that the classifier sees the lesion border and some surrounding skin. Panel D of the comparative analysis (Table VIII) supports the attention U-Net as the most accurate U-Net variant that remains light enough to train on a single GPU.

D. Hybrid Backbone Ensemble

Three backbones pretrained on ImageNet-21k (public `timm` weights) are fine-tuned with a new eight-class head:

- **EfficientNetV2-S** [71]: compound-scaled CNN; EfficientNets are the most successful family in ISIC challenges [8], [39].
- **ConvNeXt-T** [34]: a modernized CNN whose ensemble gave the best reported ISIC 2019 sensitivity [35].
- **Swin-T** [32]: a hierarchical transformer that captures global context and makes errors that differ from those of the CNNs [33].

Diversity is deliberate: combining local (CNN) and global (transformer) features consistently improves results over either alone [36], [37].

E. Metadata Fusion (Next Stage)

Age, sex, and anatomical site are available in ISIC 2019/2020, MILK10k, and PAD-UFES-20. They will be one-hot or min–max encoded and passed through a two-layer perceptron (MLP) $g(\cdot)$. For backbone m with pooled image features $\mathbf{f}_m$, the fused representation is

$$\mathbf{z}_m = \big[\, \mathbf{f}_m \,\|\, g(\mathbf{x}_{meta}) \,\big], \qquad \mathbf{p}_m = \mathrm{softmax}(W_m \mathbf{z}_m + \mathbf{b}_m), \tag{2}$$

where $\|$ denotes concatenation. Metadata gave consistent gains in the ISIC 2019 winning solution [8].

F. Imbalance-Aware Training

Each backbone is trained with the multi-class focal loss [46]

$$\mathcal{L}_{FL} = -\sum_{c=1}^{C} \alpha_c \,(1 - p_c)^{\gamma}\, y_c \log p_c, \tag{3}$$

where y_c is the one-hot label, p_c the predicted probability, $\gamma = 2$ the focusing parameter, and $\alpha_c \propto 1/\sqrt{n_c}$ a class weight computed from the number of training images n_c of class c and normalized to mean one. Augmentation (horizontal and vertical flips, random rotation up to 45°, random resized crops, and colour jitter) further protects the minority classes SCC, AK, and DF; no re-sampling is used.

Algorithm 1 Training and evaluation of the hybrid ensemble

Require: ISIC 2019 images and labels; PAD-UFES-20 (external)
Resize images (shorter side 256 px); compute a 64-bit image hash
Group images by lesion ID and identical hash
Stratified group split into train / val / test (≈71/14/14%)
for backbone $m \in$ {EffNetV2-S, ConvNeXt-T, Swin-T} **do**
    Initialize from ImageNet-21k weights; new 8-class head
    for epoch $= 1$ to 20 **do**
        Train with focal loss (Eq. 3), AdamW, warm-up + cosine
        Keep checkpoint with best validation balanced accuracy
        Stop if no improvement for 6 epochs
    end for
end for
Fit temperature τ_m per model on the validation set
Choose weights w_m (equal, or grid search on validation; Eq. 4)
Choose threshold θ for $\geq 95\%$ malignant sensitivity on validation
return evaluate once on the internal test set and on PAD-UFES-20

G. Calibrated Ensemble and Decision Rule

The final probability vector is a weighted soft vote over the $M = 3$ models with test-time augmentation (TTA) over $T = 4$ views (original, horizontal flip, vertical flip, and both):

$$\hat{\mathbf{p}} = \sum_{m=1}^{M} w_m \,\mathrm{softmax}\Big(\frac{1}{T\tau_m}\sum_{t=1}^{T}\mathbf{s}_{m,t}\Big), \qquad \sum_m w_m = 1, \tag{4}$$

where $\mathbf{s}_{m,t}$ are the logits of model m on view t and τ_m is a temperature fitted per model by minimizing the validation negative log-likelihood. We evaluate two weightings: *equal* weights ($w_m = 1/3$) and *learned* weights chosen by a grid search (step 0.05) that maximizes validation balanced accuracy. Calibration makes the malignancy probability $P_{mal} = \sum_{c\in\{\mathrm{MEL,BCC,SCC,AK}\}} \hat{p}_c$ meaningful. A lesion is flagged for referral when P_{mal} exceeds a threshold θ selected on the validation set as the largest value that still gives at least 95% malignant sensitivity. This operating point is chosen explicitly because prospective studies show that specificity collapses when it is left implicit [9], [10].

H. Explainability (Next Stage)

For the predicted class c, Grad-CAM [51] computes channel weights and a localization map from the last convolutional (or last Swin stage) feature maps A^k:

$$\alpha_k^c = \frac{1}{Z}\sum_i\sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L^c_{\text{Grad-CAM}} = \mathrm{ReLU}\Big(\sum_k \alpha_k^c A^k\Big). \tag{5}$$

The ensemble map is the weighted average of the three model maps, and explanation quality will be quantified by the fraction of heatmap energy that falls inside the segmentation mask.

I. Training Procedure

Algorithm 1 summarizes training, and Table III lists the settings used. Each backbone was trained on a single NVIDIA GeForce RTX 3060 Laptop GPU (6 GB) with bfloat16 mixed precision; peak GPU memory stayed below 3 GB. Training took 43, 49, and 36 minutes for EfficientNetV2-S, ConvNeXt-T, and Swin-T, respectively (Swin-T stopped early after 11 epochs).

V. Experimental Design and Evaluation Metrics

A. Leakage-Free Protocol

Following the recommendations in [7], every image receives a 64-bit difference hash. Images that share a lesion identifier or an identical hash are merged into one group; the hash step merged 344 groups that would otherwise have been split apart. ISIC 2019 is then divided with stratified group k-fold sampling into training (18,076 images),

TABLE III
TRAINING CONFIGURATION USED IN THE EXPERIMENTS.

Setting	Value
Input size	224×224 (random resized crop, scale 0.5–1.0)
Optimizer	AdamW, weight decay 0.05, gradient clipping 1.0
Learning rate	10^{-4} backbone, 10^{-3} head; 1-epoch warm-up, cosine decay
Epochs / batch	up to 20 (early stopping, patience 6) / 32
Loss	Focal, $\gamma = 2$, $\alpha_c \propto 1/\sqrt{n_c}$
Regularization	Stochastic depth 0.1
Augmentation	H/V flip, rotation $\leq 45^\circ$, resized crop, colour jitter
Data split	Lesion-grouped, stratified: 18,076 / 3,659 / 3,596
TTA / ensemble	4 flips; temperature scaling; soft vote
Hardware	RTX 3060 Laptop GPU (6 GB), bfloat16
Framework	PyTorch 2.10, `timm` 1.0

validation (3,659), and internal test (3,596) sets, with no group shared between sets. The validation set is used for checkpoint selection, temperatures, ensemble weights, and the referral threshold; the internal test set and PAD-UFES-20 are used only once, after all choices were frozen. For PAD-UFES-20, the eight-class output is restricted to its six classes and renormalized.

B. Metrics

With TP_c, FN_c, FP_c, and TN_c the per-class confusion counts over C classes, we report

$$\mathrm{Acc} = \frac{\sum_c TP_c}{N}, \qquad \mathrm{BA} = \frac{1}{C}\sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c}, \tag{6}$$

$$\mathrm{Prec}_c = \frac{TP_c}{TP_c + FP_c}, \qquad \mathrm{Spec}_c = \frac{TN_c}{TN_c + FP_c}, \tag{7}$$

$$F1_c = \frac{2TP_c}{2TP_c + FP_c + FN_c}, \tag{8}$$

where N is the number of test images. We report accuracy, balanced accuracy, macro-averaged precision and F1, macro one-vs-rest ROC-AUC, malignant-versus-benign AUC, sensitivity and specificity at threshold θ, and the expected calibration error (ECE). Balanced accuracy (BA) is the primary metric because it is the official ISIC 2019 metric and is insensitive to class prevalence. 95% confidence intervals are obtained with 1,000 bootstrap resamples of the test images.

C. Fairness and Robustness Analysis

PAD-UFES-20 records the Fitzpatrick skin type for 1,494 of its images. External results are therefore also reported for Fitzpatrick groups I–II, III–IV, and V–VI. Robustness is measured as the drop from the internal to the external test set. DDI will extend the skin-tone analysis in the next stage.

D. Ablation

To measure the contribution of model combination, every subset of the three backbones is evaluated with equal-weight soft voting (Table VI). Comparisons with further baselines (ResNet-50, DenseNet-121, MobileNetV2, and a DINOv2 linear probe [53]) are planned.

VI. EXPERIMENTAL RESULTS

All numbers in this section were measured by us on held-out data that was not used for training or model selection.

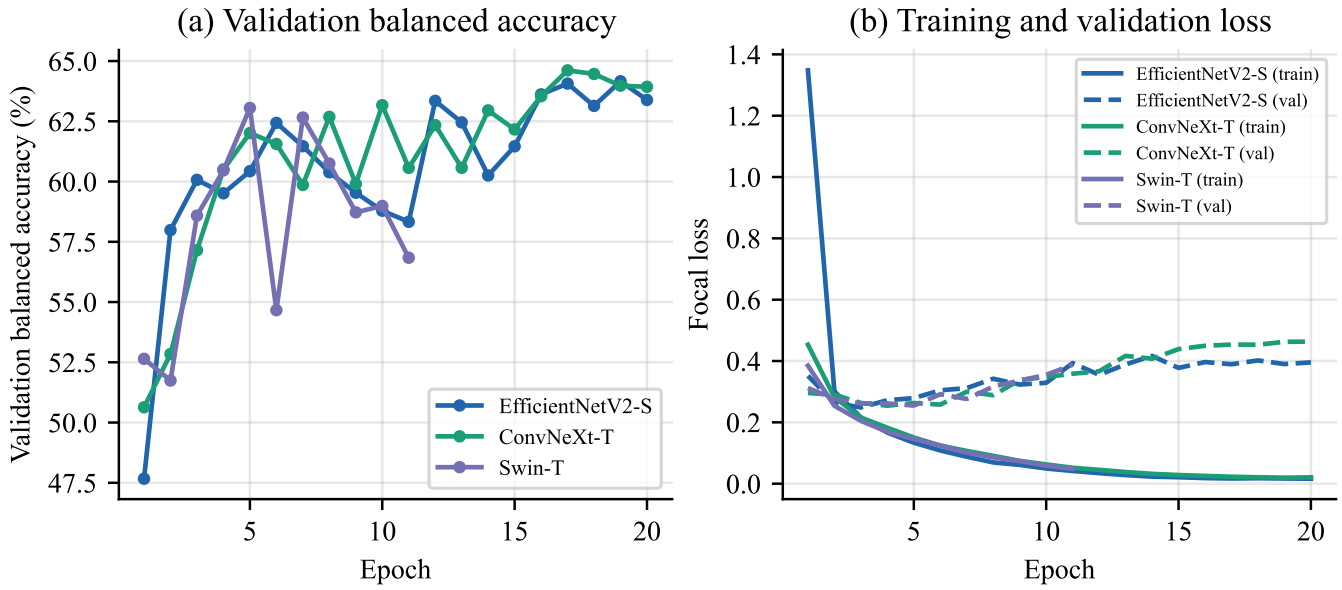


Fig. 4. Measured training curves of the three backbones on ISIC 2019: (a) validation balanced accuracy per epoch and (b) training (solid) and validation (dashed) focal loss.

TABLE IV
MEASURED RESULTS ON THE INTERNAL TEST SET (ISIC 2019, 3,596 LESION-DISJOINT IMAGES, 8 CLASSES). ACC., BA, MACRO-PRECISION (PREC.), MACRO-F1, AND MALIGNANT SENSITIVITY/SPECIFICITY IN %.

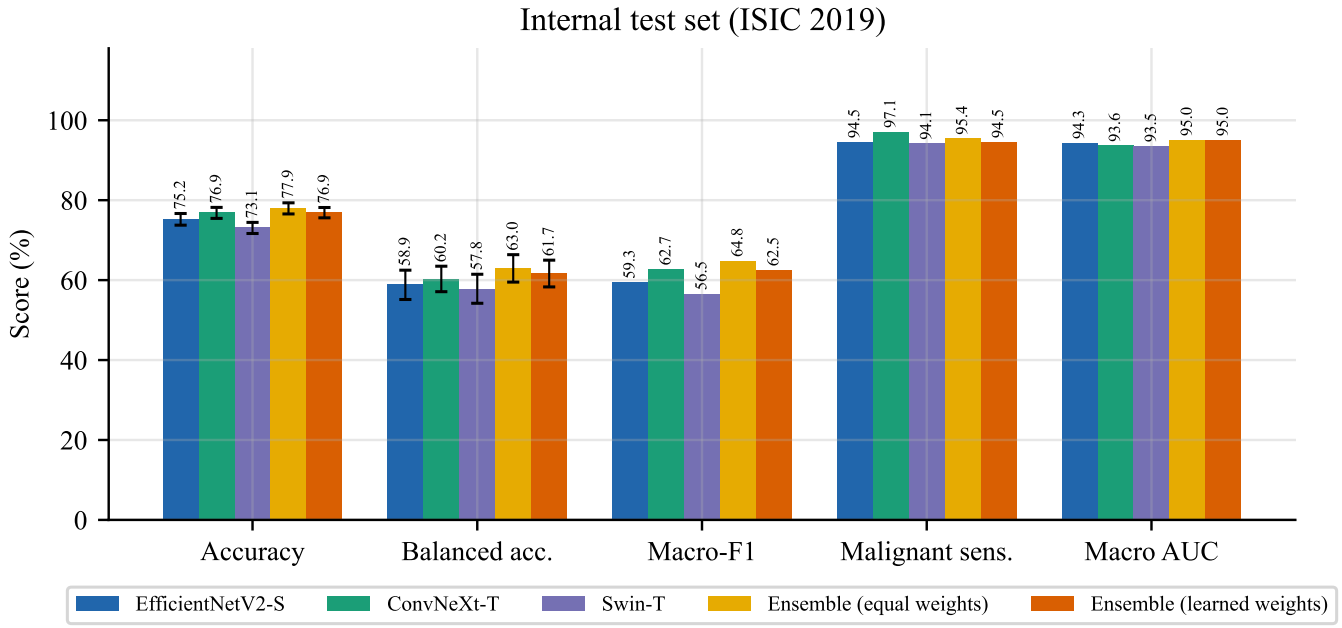
Model	Acc.	BA	Prec.	F1	AUC	Sens.	Spec.
EfficientNetV2-S	75.19	58.88	61.66	59.32	0.943	94.5	64.8
ConvNeXt-T	76.89	60.21	67.95	62.68	0.936	97.1	58.1
Swin-T	73.08	57.80	57.84	56.47	0.935	94.1	61.3
Ensemble (equal)	**77.95**	**62.99**	**68.67**	**64.79**	**0.950**	95.4	**69.9**
Ensemble (learned)	76.89	61.65	65.82	62.47	0.950	94.5	69.2

A. Training Behaviour

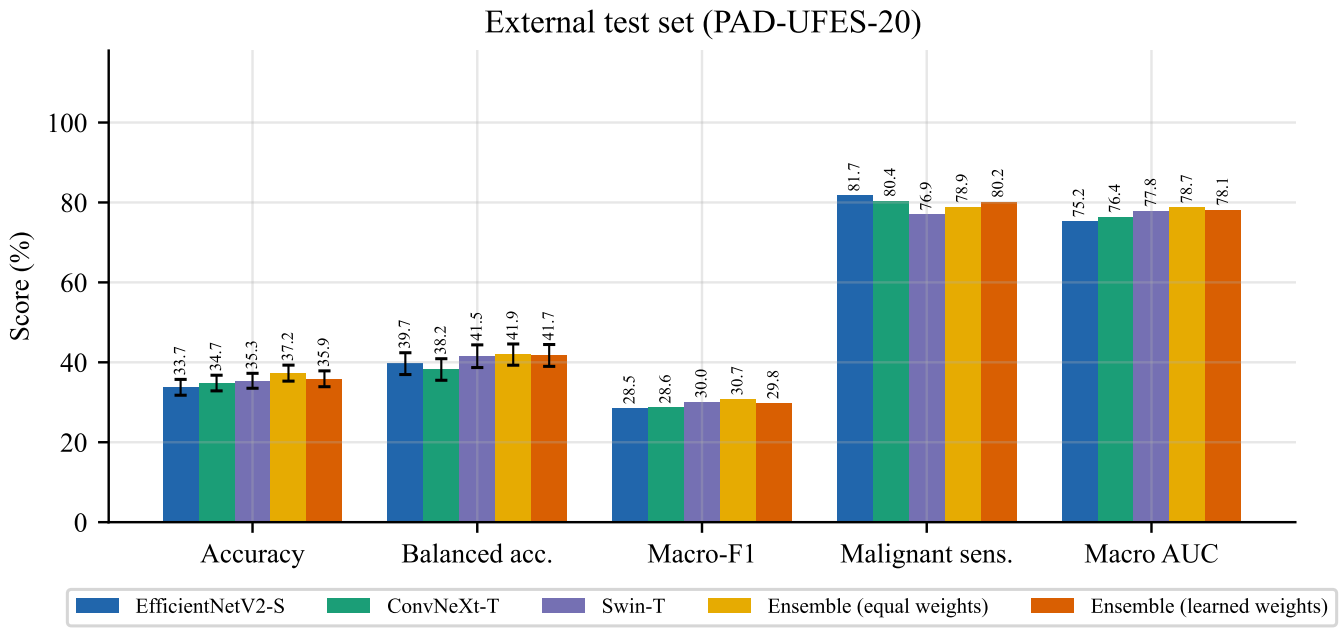
Fig. 4 shows the training curves. Validation balanced accuracy rose quickly during the first five epochs and then fluctuated between about 58% and 65%. The training loss kept falling while the validation loss began to rise after epoch 5–8, so checkpoint selection on validation balanced accuracy was essential to limit overfitting. The best validation balanced accuracies were 64.16% (EfficientNetV2-S, epoch 19), 64.61% (ConvNeXt-T, epoch 17), and 63.05% (Swin-T, epoch 5; early-stopped at epoch 11).

B. Internal Test Set (ISIC 2019)

Table IV and Fig. 5(a) report the results on the 3,596 held-out ISIC 2019 images. Among the single models, ConvNeXt-T was best (76.89% accuracy, 60.21% balanced accuracy), followed by EfficientNetV2-S (75.19%, 58.88%) and Swin-T (73.08%, 57.80%). Combining the three models with equal weights gave the best result on every main metric: **77.95% accuracy** (95% CI 76.6–79.3; 2,803 of 3,596 images correct), **62.99% balanced accuracy** (CI 59.5–66.4), **68.67% macro-precision**, 64.79% macro-F1, and a macro-AUC of 0.950. The validation-weighted ensemble (weights 0.60, 0.15, and 0.25 for EfficientNetV2-S, ConvNeXt-T, and Swin-T) reached 76.89% accuracy and 61.65% balanced accuracy. That it did not beat equal weighting indicates that the weight search slightly overfitted the validation set. At the referral threshold, the equal-weight ensemble detected 95.4% of malignant or pre-malignant lesions at 69.9% specificity. This is a higher specificity than any single model achieved at a similar sensitivity (58.1–64.8%), so it would send fewer harmless lesions for referral.



(a) Internal test set (ISIC 2019)



(b) External test set (PAD-UFES-20)

Fig. 5. Measured performance of the three single models and the two ensembles. Error bars show 95% bootstrap confidence intervals for accuracy and balanced accuracy.

C. Per-Class Behaviour

Fig. 6 shows the confusion matrix of the equal-weight ensemble with image counts, and Table V gives precision, recall, and F1 per class. Nevi (91.4% recall, 86.4% precision), vascular lesions (81.0%, 89.5%), and BCC (79.6%, 77.0%) were recognized best, followed by melanoma (65.8%, 68.5%). The hardest classes were the rare ones: SCC (48.5%, 39.3%), AK (43.0%, 38.5%), and dermatofibroma (39.0% recall at 76.7% precision). The errors are clinically plausible. Melanomas were most often confused with nevi (159 of 641), AK and SCC with BCC (45 of 128 and 16 of 68), and dermatofibromas with nevi (16 of 59). Fig. 7 shows that the ensemble also gives the best

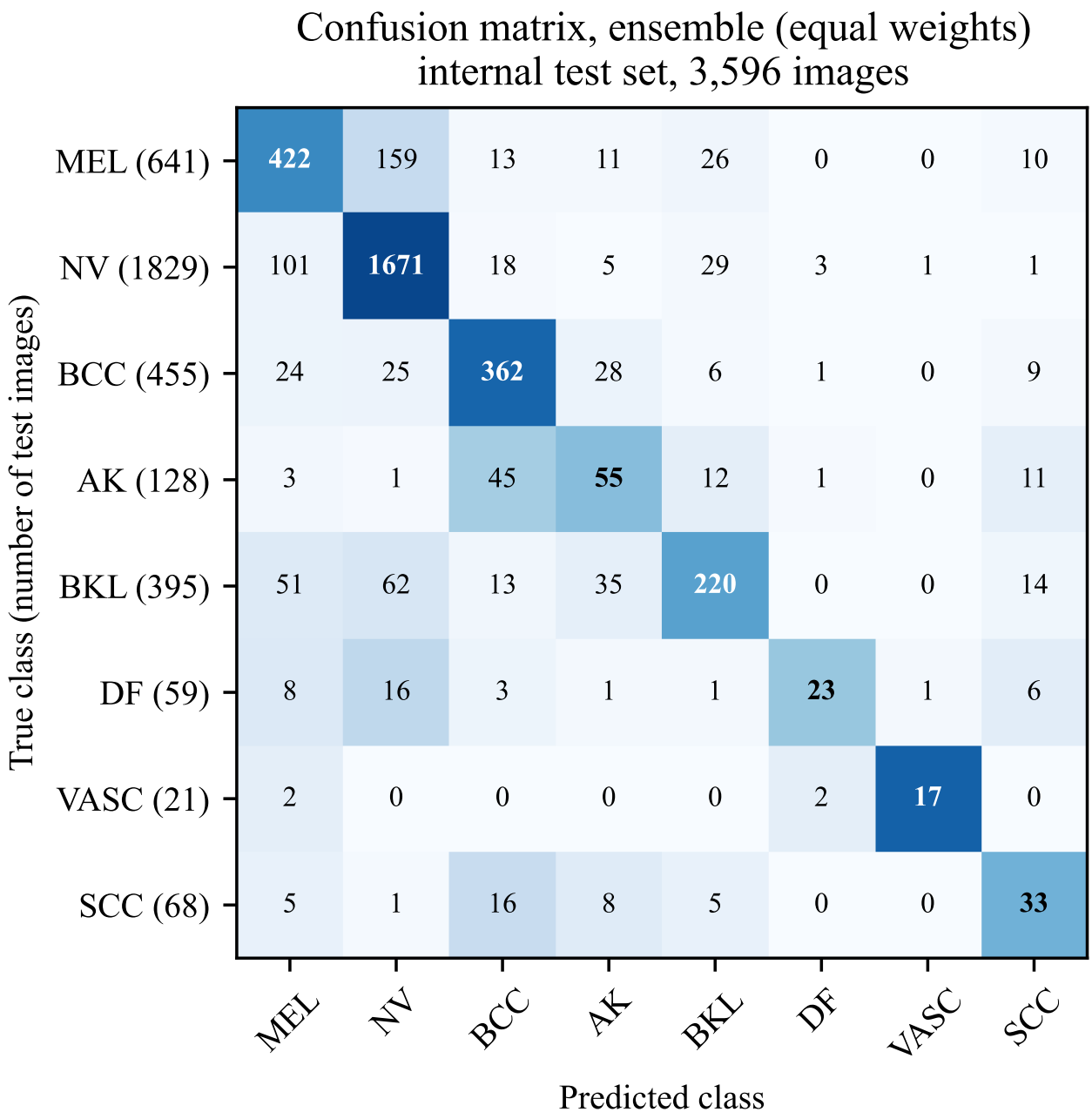


Fig. 6. Confusion matrix of the equal-weight ensemble on the internal test set (number of images; diagonal = correct). Accuracy = sum of the diagonal / all images = 2,803 / 3,596 = 77.95%.

malignant-versus-benign ROC curve (AUC 0.934 with equal weights, against 0.910–0.926 for the single models).

D. Effect of Combining Models

Table VI isolates the contribution of model combination. Every pair of backbones beat each of its members, and the full three-model ensemble beat every pair. Adding the transformer (Swin-T) to the two CNNs raised balanced accuracy from 61.31% to 62.99% and the external malignant AUC from 0.860 to 0.873, even though Swin-T was the weakest single model. This supports the central idea of this work: models with different inductive biases make different errors, and combining them improves the result.

TABLE V
PER-CLASS PRECISION, RECALL, AND F1 (%) OF THE EQUAL-WEIGHT ENSEMBLE ON THE INTERNAL TEST SET.

Class	Images	Precision	Recall	F1
Melanoma (MEL)	641	68.5	65.8	67.1
Nevus (NV)	1,829	86.4	91.4	88.8
BCC	455	77.0	79.6	78.3
Actinic keratosis (AK)	128	38.5	43.0	40.6
Benign keratosis (BKL)	395	73.6	55.7	63.4
Dermatofibroma (DF)	59	76.7	39.0	51.7
Vascular (VASC)	21	89.5	81.0	85.0
SCC	68	39.3	48.5	43.4
Macro average	3,596	68.7	63.0	64.8

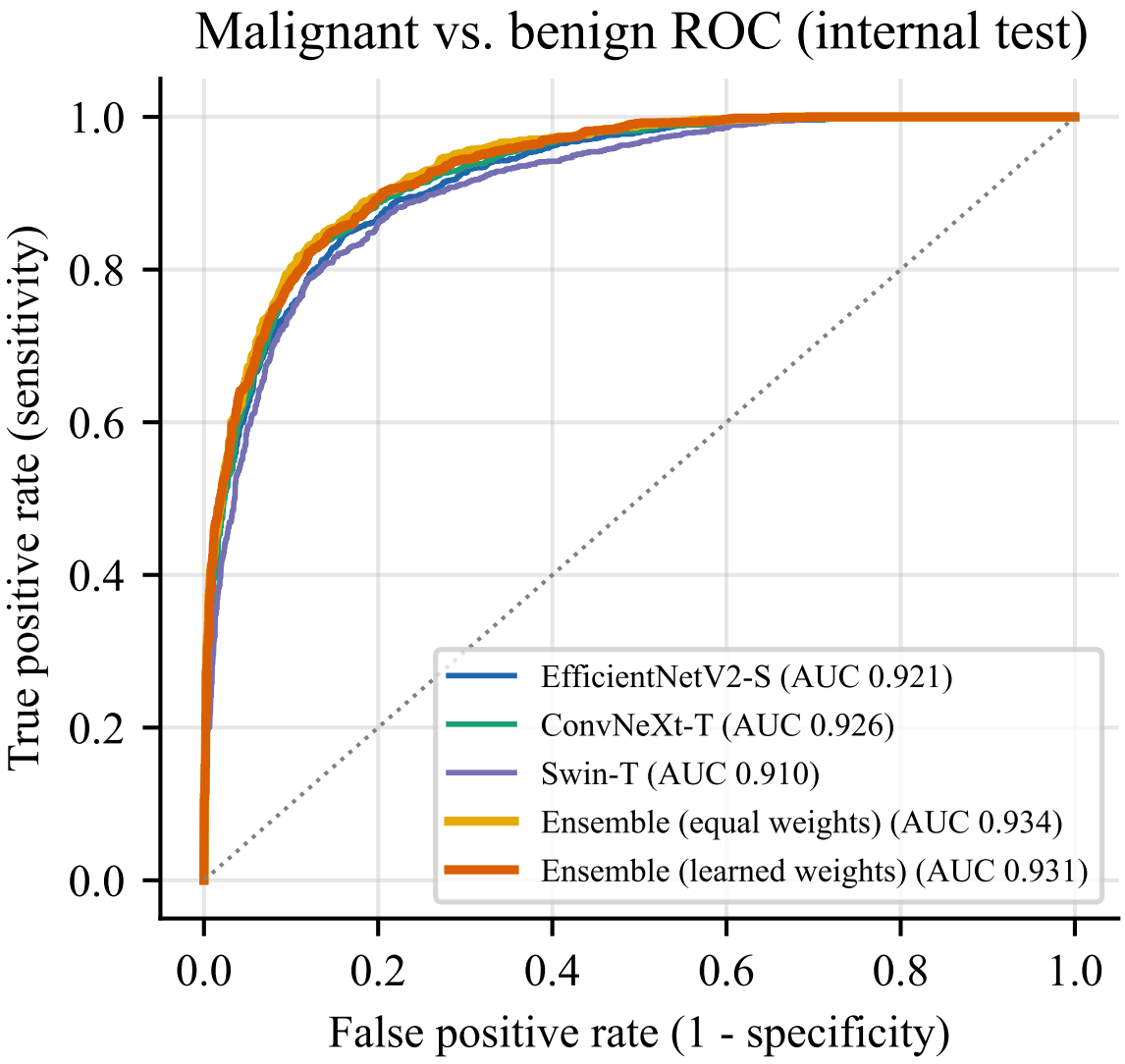


Fig. 7. Malignant (MEL, BCC, SCC, AK) versus benign ROC curves on the internal test set.

TABLE VI
ABLATION: EFFECT OF COMBINING BACKBONES (EQUAL-WEIGHT SOFT VOTING; MEASURED). INT. = INTERNAL TEST, EXT. = PAD-UFES-20; MAL. AUC = MALIGNANT-VERSUS-BENIGN AUC.

Models combined	**Int. Acc.**	**Int. BA**	**Int. Mal. AUC**	**Ext. Mal. AUC**
EffNetV2-S	75.19	58.88	0.921	0.833
ConvNeXt-T	76.89	60.21	0.926	0.844
Swin-T	73.08	57.80	0.910	0.844
EffNetV2-S + ConvNeXt-T	77.20	61.31	0.933	0.860
EffNetV2-S + Swin-T	76.70	62.39	0.928	0.864
ConvNeXt-T + Swin-T	77.09	61.55	0.929	0.866
All three	**77.95**	**62.99**	**0.934**	**0.873**

TABLE VII
MEASURED RESULTS ON THE EXTERNAL TEST SET (PAD-UFES-20, 2,298 SMARTPHONE IMAGES, 6 SHARED CLASSES).

Model	**Acc.**	**BA**	**Prec.**	**F1**	**AUC**	**Sens.**	**Spec.**
EfficientNetV2-S	33.68	39.74	34.75	28.50	0.752	81.7	68.5
ConvNeXt-T	34.73	38.22	33.82	28.64	0.764	80.4	74.3
Swin-T	35.34	41.50	35.34	30.03	0.778	76.9	75.6
Ensemble (equal)	**37.25**	**41.89**	**37.05**	**30.74**	**0.787**	78.9	**80.8**
Ensemble (learned)	35.86	41.72	35.75	29.78	0.781	80.2	77.2

E. External Test Set (PAD-UFES-20)

Table VII and Fig. 5(b) show a large drop on smartphone images. Accuracy fell to 33.7–37.3% and balanced accuracy to 38.2–41.9% for all models. Per class, nevi (82% recall) and benign keratoses (68%) still transferred reasonably, but BCC (43%), melanoma (37%), AK (14%), and SCC (8%) did not. Two factors explain this. First, PAD-UFES-20 consists of clinical photographs taken with ordinary phones, whereas all training images are dermoscopic. Second, its class mix differs strongly (37% BCC and 32% AK). The ranking of lesions by malignancy is more robust than the eight-class decision. Combining the three models raised the malignant-versus-benign AUC from 0.833–0.844 (single models) to **0.873**. At the threshold chosen on internal validation data, the ensemble detected 78.9% of malignant or pre-malignant lesions at 80.8% specificity. These results reproduce, with our own models, the benchmark-to-clinic gap documented in the literature [4], [11].

F. Skin-Type Analysis

On PAD-UFES-20, the equal-weight ensemble reached a malignant AUC of 0.868 for Fitzpatrick types I–II (1,029 images) and 0.845 for types III–IV (454 images), with similar accuracy (35.3% and 36.3%). Only 11 images belong to types V–VI, which is far too few for a reliable estimate. This confirms the diversity gap reported in the literature [12] and is why the DDI dataset is planned as the next fairness test.

VII. COMPARATIVE ANALYSIS OF ALGORITHMS

This section compares algorithm families using results reported in the literature, and places our own measured result alongside them. Because numbers from different protocols are not directly comparable, results are grouped into four protocol-consistent panels: (A) HAM10000/ISIC 2018 seven-class with random splits, (B) ISIC 2019 eight-class, (C) patient-disjoint HAM10000 with frozen encoders, and (D) segmentation on ISIC 2018 under one re-implementation. Table VIII lists all values with their protocol.

A. Panel A: Seven-Class HAM10000 / ISIC 2018

Fig. 8 ranks eleven algorithms. Deep models clearly outperform the classical baselines, and within deep models the trend moves from plain CNNs (74–88%) through attention-augmented CNNs and hybrids (89–94%) to transformers

TABLE VIII
COMPARATIVE ANALYSIS OF ALGORITHMS FOR SKIN LESION CLASSIFICATION AND SEGMENTATION (VALUES EXACTLY AS REPORTED IN EACH SOURCE).

Panel	Family	Model	Dataset (classes)	Reported result	Protocol / note
A	Classical ML	SVM [29]	HAM10000 subset (7)	Acc. 53%, W-F1 0.50	100 img/class + aug., k-fold
	Classical ML	Random forest [29]	HAM10000 subset (7)	Acc. 87%	Same protocol as SVM
	CNN	ResNet-50, CE loss [36]	ISIC 2018 (7)	Acc. 74.2%	Random 70/10/20
	CNN (light)	MobileNet [24]	HAM10000 (7)	Acc. 83.1%, W-F1 0.83	Random split
	CNN	ResNet-50, focal loss [36]	ISIC 2018 (7)	Acc. 83.7%	Same split as CE row
	CNN	EfficientNet-B4 [28]	HAM10000 (7)	Acc. 87.9%, F1 0.87	Best of B0–B7
	Hybrid	ResNet + ViT, focal [36]	ISIC 2018 (7)	Acc. 89.5%, macro-F1 81.1%	Same split as CE row
	CNN	ResNeXt-101 [25]	HAM10000 (7)	Acc. 93.2%	Best of five CNNs
	CNN + attention	InceptionResNetV2 + SA [26]	HAM10000 (7)	Acc. 93.4%, AUC 0.984	Resampled random split
	Hybrid	Swin + DenseNet attn. [37]	HAM10000 (7)	Acc. 94.2%	Single dataset
	Transformer	SkinTrans (ViT) [31]	HAM10000 (7)	Acc. 94.3%	Split not detailed
B	CNN	EfficientNet-B0 [8]	ISIC 2019 (8)	BA 65.8% (67.4% + meta)	5-fold CV
	Ensemble	Multi-res. EfficientNets + meta [8]	ISIC 2019 (8)	BA 74.2% (CV)	5-fold CV
	Ensemble	same, official test [8]	ISIC 2019 (8+unk.)	BA 63.6%	Hidden test with unknown class
	Transformer	Swin [33]	ISIC 2019 (8)	BA 82.3%, spec. 97.9%	Hold-out split
	Ensemble	ConvNeXt T/S/B/L [35]	ISIC 2019 (8)	Sens. 84.2%, spec. 97.9%	Same group as Swin row
	Ensemble	**EffNetV2-S + ConvNeXt-T + Swin-T (ours)**	ISIC 2019 (8)	**BA 63.0%, acc. 78.0%**	Lesion-disjoint held-out test, 224 px
C	ViT (ImageNet)	ViT-L linear probe [53]	HAM10000 clean (7)	BA 65.3%, AUC 0.970	Patient-disjoint 60/20/20
	Self-supervised	DINOv2 ViT-L probe [53]	HAM10000 clean (7)	BA 70.1%, AUC 0.964	Patient-disjoint
	Foundation	PanDerm ViT-L probe [53]	HAM10000 clean (7)	BA 80.7%, AUC 0.988	Patient-disjoint
–	Ensemble	EfficientNet ens. + meta [39]	ISIC 2020 (2)	AUC 0.949	Official private leaderboard
	CNNs (19)	Best after de-dup. [7]	ISIC 2020 (2)	AUC 0.80	Duplicates removed
D	CNN	U-Net [45]	ISIC 2018 T1	Dice 0.867, IoU 0.849	Uniform re-run, 224 px
	CNN + attention	Attention U-Net [45]	ISIC 2018 T1	Dice 0.882, IoU 0.865	Same
	Transformer	TransUNet [45]	ISIC 2018 T1	Dice 0.850, IoU 0.837	Same
	Hybrid	UCTransNet [45]	ISIC 2018 T1	Dice 0.890, IoU 0.873	Same

(94%). The most informative comparison is the controlled one by Nie *et al.* [36]: with backbone and split fixed, focal loss adds 9.5 points and the hybrid transformer adds another 5.7. The *training strategy* can therefore matter as much as the architecture. Note that all Panel A results use image-level random splits, which are known to be optimistic [7].

B. Panel B: Eight-Class ISIC 2019

ISIC 2019 is harder: it adds SCC, has a larger and more heterogeneous image pool, and is normally scored with balanced accuracy. Fig. 9 shows two lessons. First, ensembles beat single models (74.2% vs. 65.8% BA for Gessert *et al.*), and modern backbones (Swin, ConvNeXt) score higher still. Second, the same winning ensemble dropped from 74.2% in cross-validation to 63.6% on the official hidden test set, which also contains unseen "unknown" lesions. This ten-point drop is larger than the differences between most architectures, and it is the main reason our protocol evaluates on external data. Our ensemble reached 63.0% balanced accuracy on a lesion-disjoint held-out test set (orange bar). That is the same level as the ISIC 2019 winner on the official test set (63.6%), although the

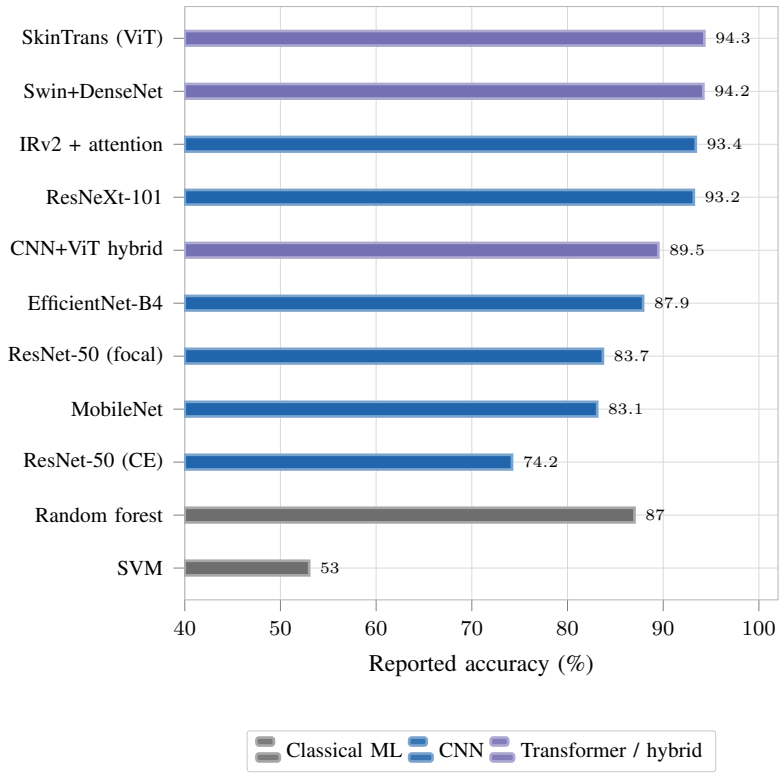


Fig. 8. Panel A: reported seven-class accuracy on HAM10000/ISIC 2018 (random splits) for classical, CNN, and transformer/hybrid algorithms (sources in Table VIII).

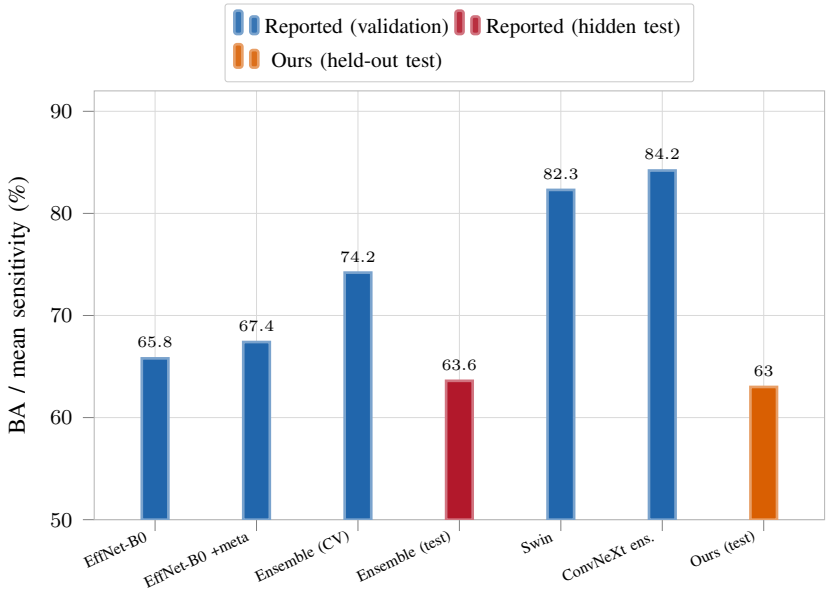


Fig. 9. Panel B: ISIC 2019 eight-class balanced accuracy. The red bar shows the ISIC 2019 winner on the official test set, 10.6 points below its own cross-validation score [8], [33], [35]; the orange bar is our measured ensemble on a lesion-disjoint test set.

two test sets differ and that system used higher resolutions and metadata. It is clearly below the hold-out results reported for Swin and ConvNeXt ensembles, whose split details were not reported.

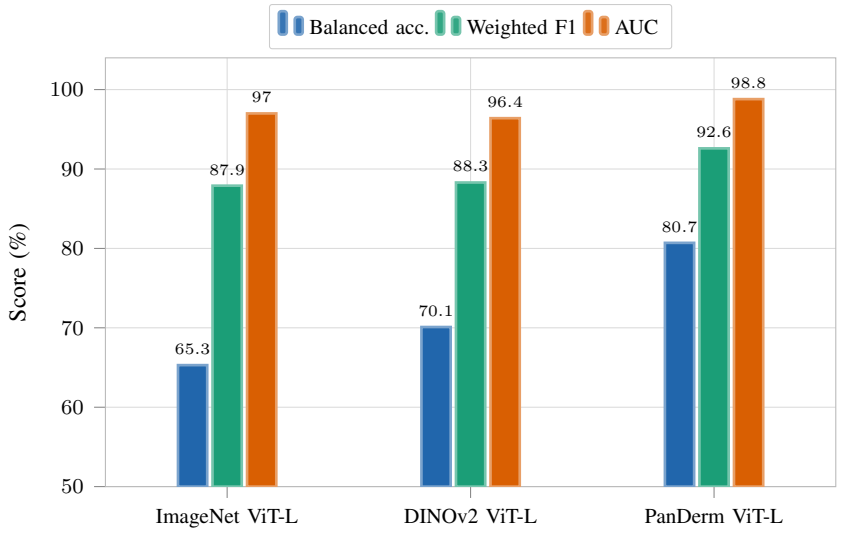


Fig. 10. Panel C: linear-probe results on a patient-disjoint HAM10000 split [53]. Domain-specific pretraining gives the largest balanced-accuracy gain.

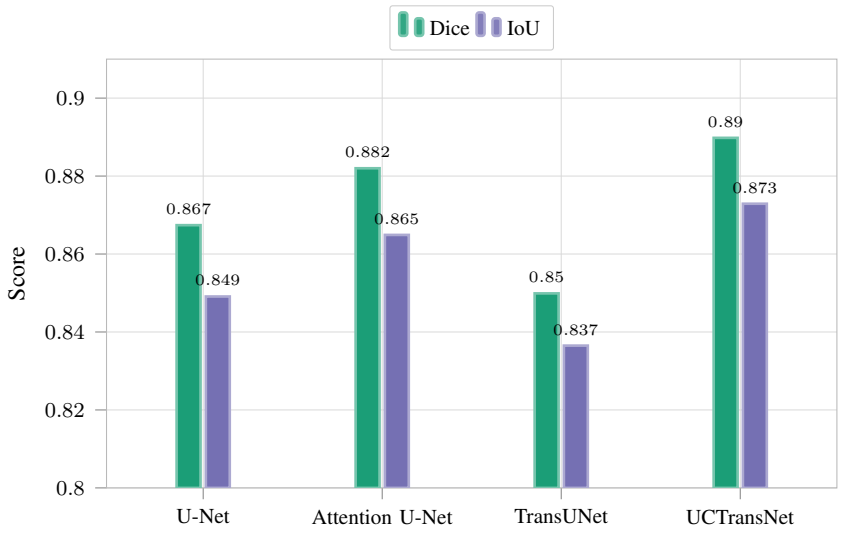


Fig. 11. Panel D: segmentation performance on ISIC 2018 Task 1 under a uniform protocol [45].

C. Panel C: Patient-Disjoint Evaluation of Pretraining

Panel C is the fairest comparison, because all three encoders were evaluated by the same authors with a patient-disjoint split and an identical linear classifier [53]. Fig. 10 shows that the *pretraining source* changes balanced accuracy by more than 15 points: ImageNet features reach 65.3%, general self-supervised DINOv2 features 70.1%, and dermatology-specific PanDerm features 80.7%. AUC hides this difference (0.96–0.99), which again argues for balanced accuracy as the primary metric.

D. Panel D: Lesion Segmentation

Under a single uniform re-implementation on ISIC 2018 [45], attention gates improve U-Net Dice from 0.867 to 0.882, and the hybrid UCTransNet reaches 0.890, whereas the pure-transformer TransUNet is lowest at 0.850 (Fig. 11). With limited data, the convolutional inductive bias still helps; this supports our choice of an attention U-Net for the segmentation stage.

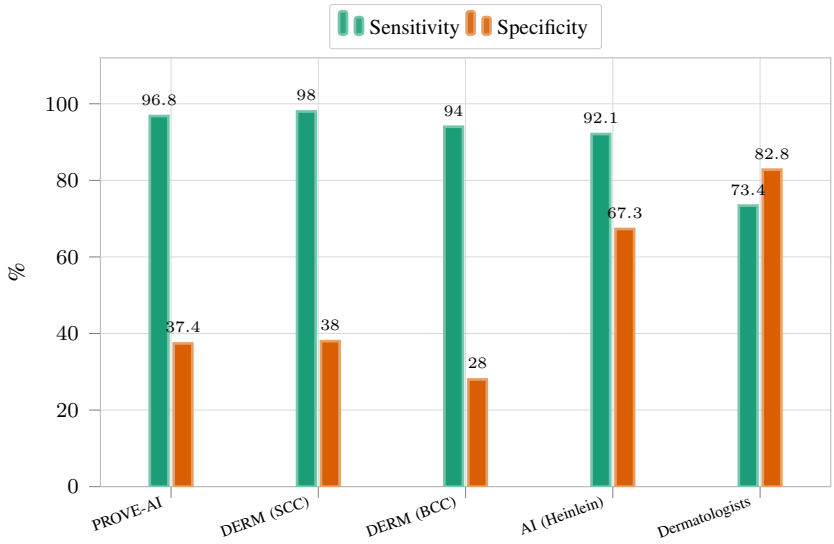


Fig. 12. Prospective clinical performance: AI systems achieve high sensitivity at low specificity [9], [10], [58].

E. Benchmark versus Clinic

Fig. 12 compares sensitivity and specificity from three prospective studies. All AI systems operate at high sensitivity but with specificities of 28–67%, which would generate many unnecessary referrals or biopsies. Dermatologists in the same multicentre study had lower sensitivity but much higher specificity [58]. These findings motivate the explicit, validated operating threshold and calibration step in our method.

F. Qualitative Comparison of Algorithm Families

Table IX summarizes the trade-offs between families, including approximate model size, which determines whether a model can run on a phone or needs a server.

VIII. Discussion

Main findings. The comparative analysis supports four conclusions. (i) Deep learning clearly outperforms classical machine learning for skin cancer detection. (ii) Among deep models, ensembles and CNN–transformer combinations are consistently the strongest, but the margins between modern backbones are only a few points. (iii) Evaluation choices—random versus grouped splits, internal versus external data, accuracy versus balanced accuracy—shift results by 10–15 points, more than the architecture does. (iv) Imbalance handling and domain-specific pretraining are the two cheapest sources of large gains. Our own experiments confirm the second and third points. The three-model ensemble outperformed every single model and every pair on internal data (77.95% accuracy, 62.99% balanced accuracy). The same models lost about 40 points of accuracy on external smartphone images, far more than the fewer than 4 points that separate the backbones.

Why not claim 99% accuracy? Several recent papers report 97–100% accuracy on HAM10000 [38], [47]. Without patient-level splits and external tests, such numbers mainly reflect near-duplicate images. We therefore set our success criteria on external and per-skin-tone performance rather than on internal accuracy.

Why does the ensemble help? The single models make partly different errors. For example, Swin-T had the best AK recall (50%) but the worst BCC recall (65%), while ConvNeXt-T showed the opposite pattern. Averaging calibrated probabilities therefore corrects some errors of each member. It also improves specificity at a fixed sensitivity, which matters for referral. The gains are consistent but modest (about 1–3 points of balanced accuracy), and the confidence intervals of the best single model and the ensemble overlap. A larger test set or repeated splits are needed to confirm the size of the improvement.

Limitations. The reported experiments use one lesion-grouped split, 224×224 inputs, and whole images. Segmentation, metadata fusion, colour constancy, and Grad-CAM are not yet included, and the weights learned on validation data did not beat simple averaging. The external accuracy on smartphone images is low, so the

TABLE IX
QUALITATIVE COMPARISON OF ALGORITHM FAMILIES FOR SKIN CANCER DETECTION (PARAMETER COUNTS ARE THOSE OF THE STANDARD IMAGENET MODELS).

Family	Representative	Params	Strengths	Weaknesses	Used here
Classical ML	SVM, random forest	<1M	Fast, interpretable features	Far lower accuracy; handcrafted features	Baseline only
Plain CNN	VGG16 / ResNet-50	138M / 25.6M	Mature, easy transfer learning	VGG is heavy; limited global context	Baseline
Dense / Inception CNN	DenseNet-121, InceptionV3	8.0M / 23.8M	Feature reuse; multi-scale filters	Moderate accuracy on ISIC 2019	Baseline
Lightweight CNN	MobileNetV2	3.5M	Runs on smartphones	Lower accuracy on rare classes	Baseline, future distillation
Scaled CNN	EfficientNet / V2-S	5.3M–22M	Best accuracy per FLOP; challenge winner	Sensitive to input resolution	✓
Modern CNN	ConvNeXt-T	28M	Transformer-style design, CNN efficiency	Needs strong augmentation	✓
Vision transformer	ViT-B/16	86M	Global attention	Data-hungry; weak with small data	Probe baseline
Hierarchical transformer	Swin-T	28M	Global + local windows; strong on ISIC 2019	Higher memory at high resolution	✓
Hybrid CNN–ViT	ResNet+ViT, Swin+DenseNet	varies	Complementary local/global features	More complex; rarely externally tested	Ensemble design
Ensemble	Multi-backbone voting	Sum of members	Most robust; wins challenges	Slower inference	✓
Foundation model	PanDerm, DINOv2	300M+	Best transfer, label-efficient	Heavy compute; licence constraints	Future work

current models must not be used on such images without domain adaptation or fine-tuning on clinical photographs. Published numbers in Table VIII come from different authors and are comparable only within a panel. Only 11 external images come from Fitzpatrick types V–VI, so fairness for darker skin remains untested. The system is intended as a decision-support tool for clinicians, not an autonomous diagnostic device. Clinical use would require prospective validation and regulatory approval.

Ethics. Only de-identified public datasets are used, under their respective licences (mostly CC BY-NC). The model will be released with a model card that lists per-skin-tone performance and known failure modes.

IX. CONCLUSION AND FUTURE WORK

This paper reviewed deep learning for skin cancer detection, analysed eleven public datasets, and compared classical, CNN, transformer, hybrid, ensemble, foundation, and segmentation models using published results grouped by comparable protocol. The comparison shows that ensembles of complementary CNN and transformer backbones, trained with imbalance-aware losses and metadata, are the most effective practical choice. It also shows that leakage-free and external evaluation matter more than any single architecture. Building on these findings, we trained two CNNs (EfficientNetV2-S, ConvNeXt-T) and a transformer (Swin-T) on ISIC 2019 with focal loss and a leakage-free split, and combined them by calibrated soft voting. On 3,596 held-out images the ensemble achieved **77.95% accuracy**, **62.99% balanced accuracy**, a macro-AUC of **0.950**, and **95.4%** malignant sensitivity at 69.9%

specificity, outperforming every single model and every pair of models. On the external smartphone dataset PAD-UFES-20, accuracy dropped to 37.25%, but the ensemble still gave the best malignant-versus-benign AUC (0.873). Combining multiple algorithms is therefore a reliable way to improve skin cancer detection, while transfer to new imaging conditions remains the main open problem. Future work will (1) add the segmentation, metadata, and Grad-CAM modules and measure their contribution, (2) train at higher resolution and fine-tune on clinical and smartphone images (PAD-UFES-20, MILK10k) to close the domain gap, (3) evaluate fairness on DDI, (4) distil the ensemble into a MobileNet-class student for smartphone use, and (5) explore dermatology foundation-model features.

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